

Program-1: Build a Docker Container from a Custom Dockerfile

Step 1: Create Project Directory

```
mkdir Program-1  
cd Program-1
```

Step 2: Create Dockerfile

Create a file named **Dockerfile** (no extension):

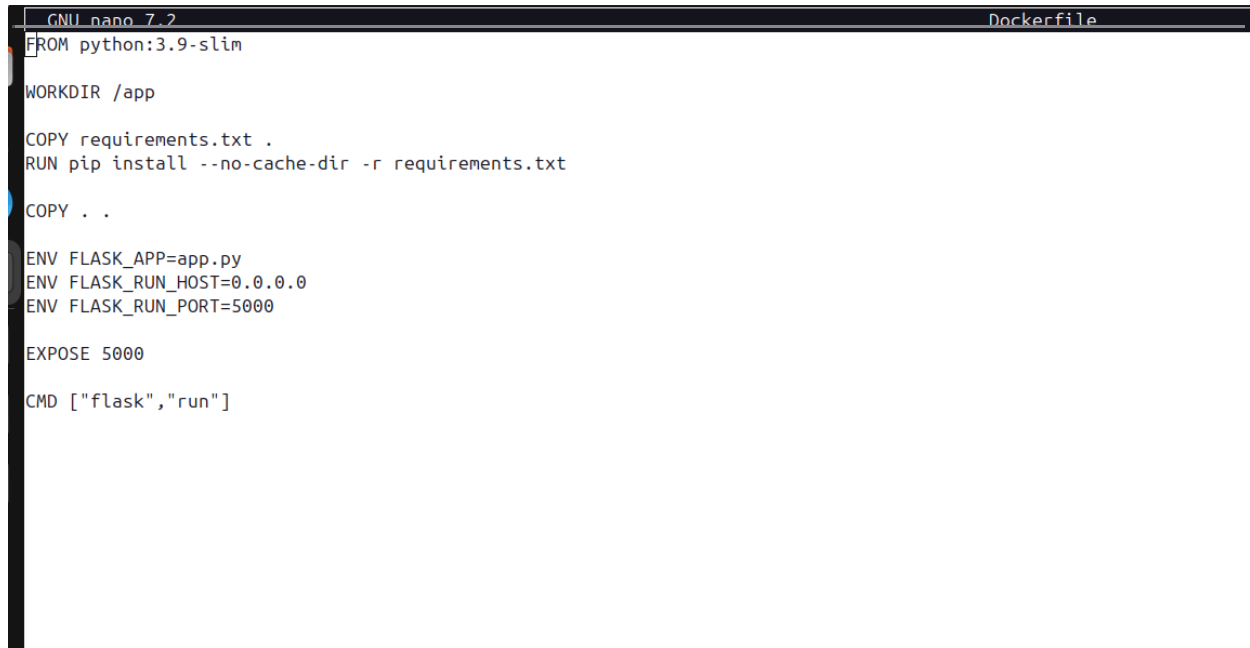
```
nano Dockerfile
```

Add the following content:

```
# Use a slim Python base image  
FROM python:3.9-slim  
  
# Set working directory  
WORKDIR /app  
  
# Copy requirements file and install dependencies  
COPY requirements.txt .  
RUN pip install --no-cache-dir -r requirements.txt  
  
# Copy application code  
COPY . .  
  
# Set Flask environment variables  
ENV FLASK_APP=app.py  
ENV FLASK_RUN_HOST=0.0.0.0  
ENV FLASK_RUN_PORT=5000  
  
# Expose Flask port  
EXPOSE 5000
```

Run the Flask application
CMD ["flask", "run"]

Save and exit (CTRL+O, ENTER, CTRL+X).

A screenshot of a terminal window showing a Dockerfile being edited with the nano text editor. The title bar of the editor shows 'GNU nano 7.2' and 'Dockerfile'. The content of the file is as follows:

```
FROM python:3.9-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

EXPOSE 5000

CMD ["flask", "run"]
```

Step 3: Create Python Application File

Create **app.py**:

```
nano app.py
```

Add the following code:

```
from flask import Flask
```

```
app = Flask(__name__)
```

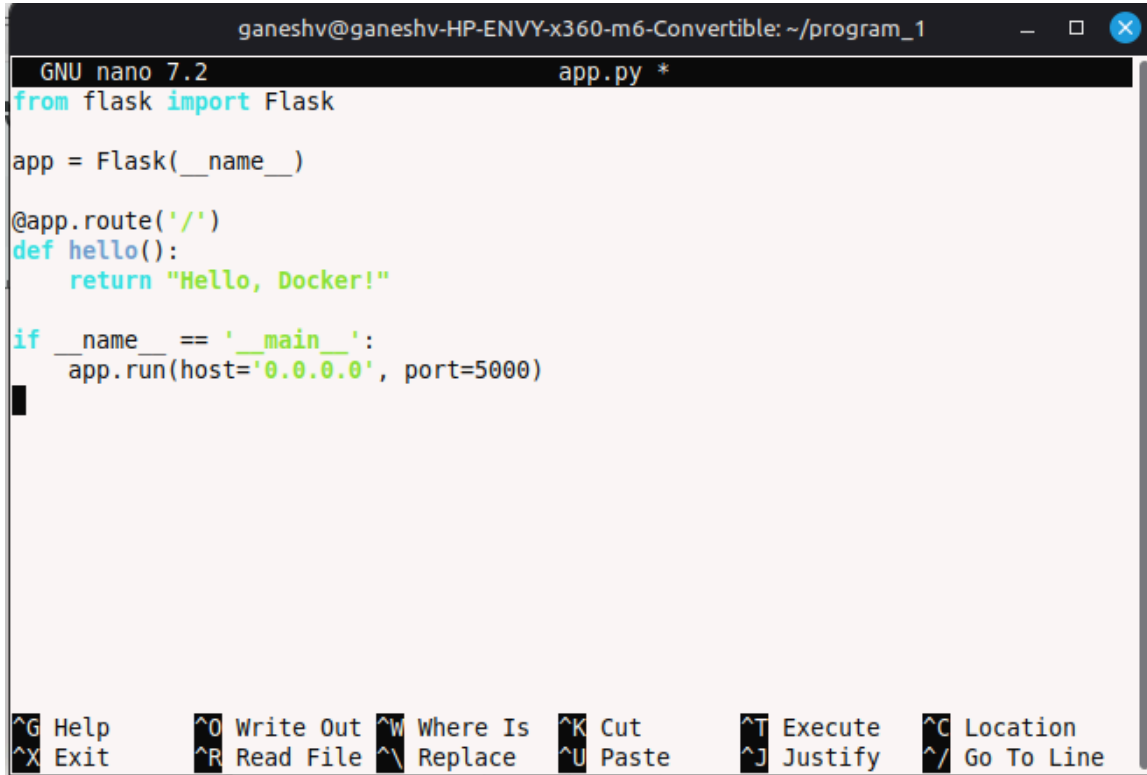
```
@app.route("/")
```

```
def hello():
```

```
    return "Hello, Docker!"
```

```
if __name__ == "__main__":  
    app.run(host="0.0.0.0", port=5000)
```

Save and exit.



```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible: ~/program_1  
GNU nano 7.2 app.py *  
from flask import Flask  
  
app = Flask(__name__)  
  
@app.route('/')  
def hello():  
    return "Hello, Docker!"  
  
if __name__ == '__main__':  
    app.run(host='0.0.0.0', port=5000)  
█  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Step 4: Create requirements.txt

nano requirements.txt

Add:

flask

Save and exit.

Step 5: Build Docker Image

Run the Docker build command:

```
docker build -t program-1 .
```

```

ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/program_1$ docker build -t program_1 .
[+] Building 14.8s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 204B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 3.7s
=> [auth] library/python:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f1 4.8s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f1 0.0s
=> => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3 1.29MB / 1.29MB 1.4s
=> => sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710a 13.88MB / 13.88MB 3.9s
=> => sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb 251B / 251B 1.2s
=> => sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13 10.36kB / 10.36kB 0.0s
=> => sha256:dad5b29e3506c35e0fd222736f4d4ef25d21b219acdd73f7bb41d5999 1.74kB / 1.74kB 0.0s
=> => sha256:085da638e1b8a449514c3fda83ff50a3bffa4418b050cfacd87e5722 5.40kB / 5.40kB 0.0s
=> => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df388 0.1s
=> => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c 0.8s
=> => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb5 0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 259B                                       0.0s
=> [2/5] WORKDIR /app                                             0.2s
=> [3/5] COPY requirements.txt .                                  0.0s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt      5.3s
=> [5/5] COPY app.py .                                           0.1s
=> exporting to image                                             0.5s
=> => exporting layers                                                0.4s
=> => writing image sha256:59fbdaa1938c5f792ba37104d0f371cc0234d2d556277f602d9c901de15 0.0s
=> => naming to docker.io/library/program_1                        0.0s
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/program_1$ █

```

Verify image creation:

docker images

Step 6: Run the Docker Container

docker run -d -p 5000:5000 --name flask-container program-1

Check running c

```

ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/program_1$ docker run -d -p 5000:5000 program_1
4b4fad77bb53468d8250f0b26da03dffd2b666c434d8bda530abc41a597300af
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/program_1$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS         NAMES
4b4fad77bb53   program_1                          "python app.py"         9 seconds ago  Up 9 seconds
0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
mystifying_bouman
ec2dc98e9fbe   fifthprogram-jenkins              "/usr/bin/tini -- /u..." 10 days ago    Up 23 minutes
0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
jenkins-ci
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/program_1$ █

```

containers: docker ps

Step 7: Test the Application

Open a browser and go to:

`http://localhost:5000`

Output displayed:

Hello, Docker!

Step 8: Stop the Container

`docker container stop <container-id>`

Step 9: Remove the Container

`docker container rm <container-id>`

Step 10: Remove Docker Image

`docker image rm program`

